# PetLoop

PetLoop – October 25, 2026

# Meet the Team

**Renée Amaya** - *Team Lead and Cloud-Video Infrastructure*

**Jennifer Le** - *Database & Algorithms Lead*

**Benny Pham** - *Backend Systems and Documentation Lead*

**Miranda Salazar** - *Front-End Lead and Quality Assurance*



# Problem Statement & Value Proposition

**Core Problem Space & Justification**

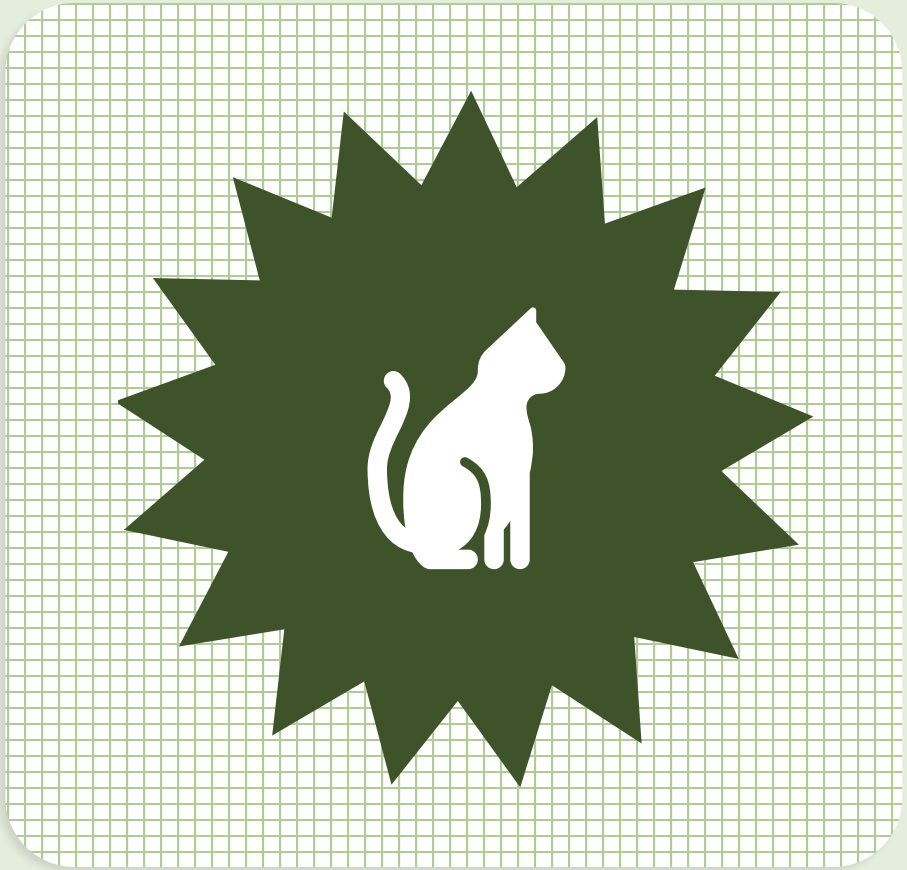
- Adoption platforms primarily reply on static images & texts.
- Currently shelters waste valuable time and labor typing out biographies and data entry forms.

**Target Audience:**

- *B2B*: Animal rescues and local rescues.
- *B2C*: Gen-Z, Millennial and other pet loving consumers.

**Intended Impact:**

- B2B Automated Intake
- B2C Swipeable Content Feed
- Overall impact shows in a decrease length of stay for animals, improved shelter costs, and reduced rehoming with better behavioral matching.



# Competitive Landscape

| | |
|---|---|
| **Petfinder/Adopt-a-Pet** | • Largest pet adoption databases.<br>• Require manual data entry, including written pet biographies and photos.<br>• Rely on traditional, filter-based searching. |
| **Good Dog** | • Screens responsible breeders and rescues.<br>• Provides a trusted environment for adopters.<br>• Still relies on traditional search and filtering methods. |
| **Pawsibly Yours** | • Offers a modern, mobile-first swipe interface.<br>• Still requires shelter staff to manually create pet listings.<br>• Has limited swipe counts and content availability. |

**KEY GAP: Manual listing work and limited discovery can slow pet visibility.**

# Tech Moat & Expected Impact

## AI-Assisted Intake (B2B)

Shelter staff upload short videos and photos. AI helps identify behavioral and physical traits to create pet profiles, reducing repetitive data entry.

## Personalized Video Feed (B2C)

Adopters browse short pet videos. The system uses watch time and swipe interactions to recommend animals based on their interests.

**EXPECTED IMPACT/ROI**

- Shelters: less administrative work and more time for animal care.
- Adopters: more pet visibility and potentially better matches, with fewer adoption returns

# Personas, Ethics & Legal Compliance

## User Personas

- Adopters/Fosters, Shelter Staff, Shelter Directors

## Ethics and Data Privacy

### Software Engineering Ethics

- Follow the ACM Code of Ethics
- Provide honest pet descriptions & accurate health information.
- Give users clear adoption guidelines.

### Data Privacy Compliance

- PetLoop collects personal information and scrolling habits to recommend suitable pets.
- GDPR: Collect only necessary data and request consent when needed.
- CCPA: Allow users to request deletion of their personal information.

## Accessibility & Intellectual Property

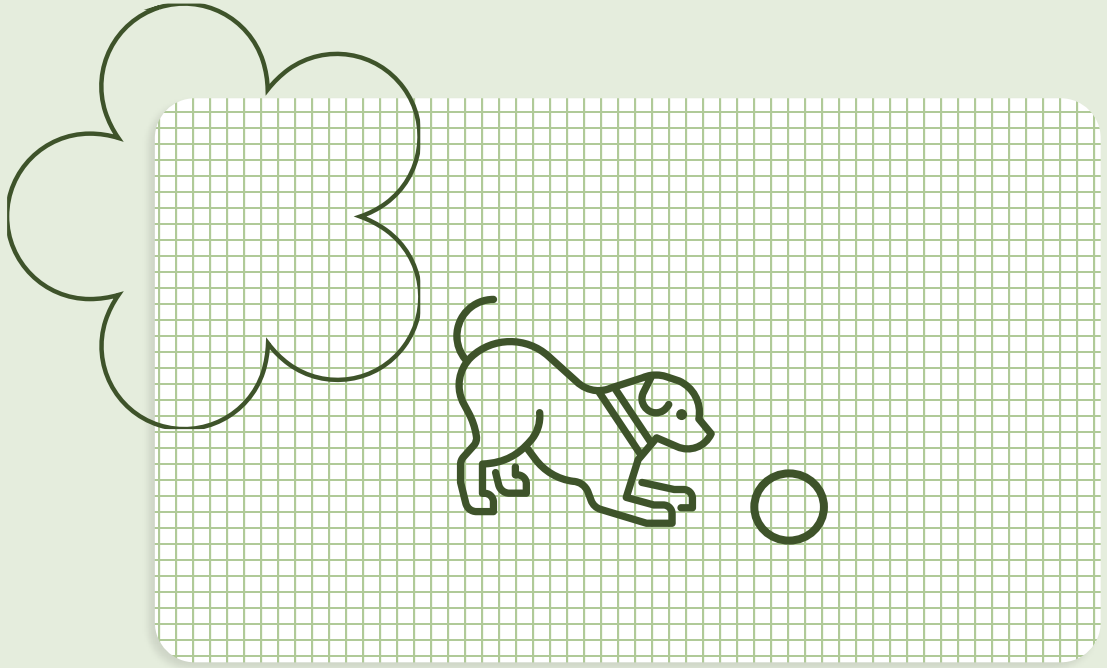
### Accessibility — WCAG 2.1

- *Perceivable*: Provide video, audio, and text information.
- *Operable*: Make searching and navigation easy.
- *Understandable*: Clearly display pet details.
- *Robust*: Support different devices, browsers, and accessibility tools.

### Intellectual Property & Licensing

- Use properly licensed datasets for personalized pet recommendations.
- *CC0*: Allows broad reuse.
- *CC BY*: Requires credit to the original creator.
- Follow licensing requirements when using open-source software.

# Final Top-Down Requirements & Test Matrix



### System Decomposition Domains

- AUTH, INGEST, AI, FEED, INTERACT, ALGO, INFRA

### MVP Scope Tags

- *Must-Have:* Core Authentication, Video Ingestion, and Discovery Feed features
- *Should-Have*: Advanced algorithm recommendations and AI safety moderation
- *Nice-to-Have:* Bonus features like administrative alerts or AI chatbot

# Final Top-Down Requirements & Test Matrix

## NFR Benchmarks

- *Performance:* Video Load Latency < 600 milliseconds (NFR-INFRA-01)
- *Security:* Secure data handling & authentication (NFR-INFRA-02)
- *Reliability:* Minute live database syncs for pet availability (NFR-INFRA-03)

## Verification Mapping

- Unit Testing & Input Inspection
- UI Inspection & Functionality Tests
- Automated Integration Testing
- Load & Benchmark Testing
- Security Audit

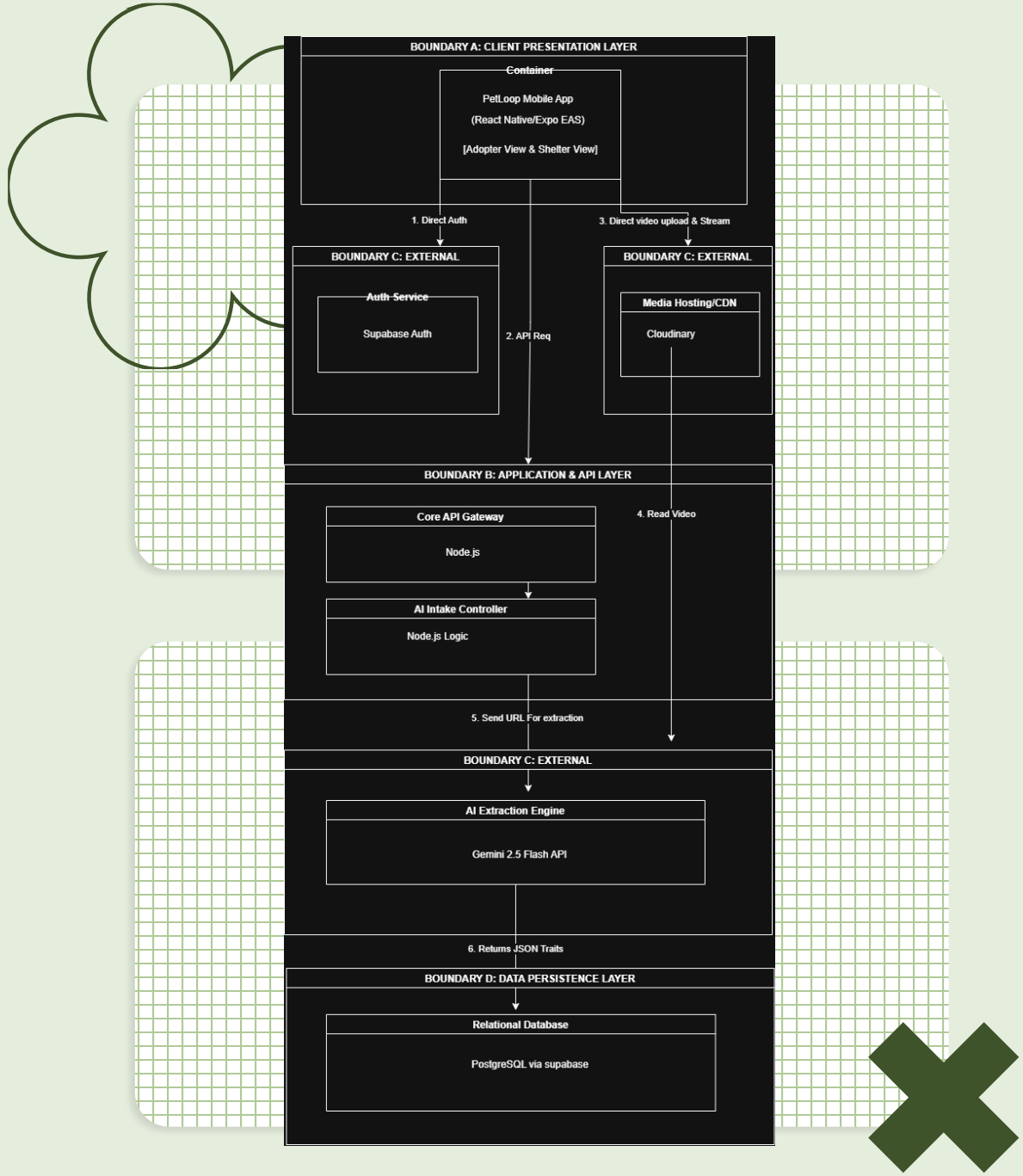
# Architectural Blueprint & System Design

## Cloud/Infrastructure Layers

- Client, Presentation Layer (Front End): React Native
- Application & API Layer: Node.js
- Ext. Cloud Services: Supabase Auth, Cloudinary, Gemini 2.5 Flash

## Technology Stack Justification

- Familiarity and Compatibility
- Cost-Effectiveness
- Simplicity
- Scalability + Performance



# Evolution of Engineering Roles

- ***Team Lead:*** Project coordination, Team planning, Communication
- ***Cloud-Video Infrastructure:*** External Cloud Services Management, Cloud-to-App Integration
- ***Database & Algorithms Lead:*** PostgreSQL Database Designing, Data Handling, Algorithm Development
- ***Backend Systems:*** Development in Node.js + TypeScript, Cross-Service Integration
- ***Documentation Lead:*** GitHub Repo Maintenance, Project Documentation
- ***Front-End Lead:*** React UI Development, App Navigation
- ***Quality Assurance:*** Functional Testing, Project Validation

# Schedule

**Team Workflow Methodologies:**

- 1 week cycle with team deliverables
- Informal check-ins ~3 times a week
  - 1: Discuss next timeline piece, split responsibilities, discuss ideas
  - 2: Ongoing check-in to discuss progression, any difficulties or review
  - 3: Finalization check with whole team
- Tracking and documentation
  - Discord with organized channels
  - Email for file sharing

**Milestones**

- M1: Project Brand
  - Target audience, role assignment, and scope
- M2: Technical Specifications
  - Tech stack, UI architecture, overall features
- M3: Feedback & Refinement
  - Peer review, adjust and rework requirements and critiques
- M4: MVP Development
  - Prototype basic functionality and minimum requirements
- M5: PetLoop Production
  - Full development – sub cycles
- M6: QA & Beta Testing
  - Internal  followed by public review
- M7: Deployment



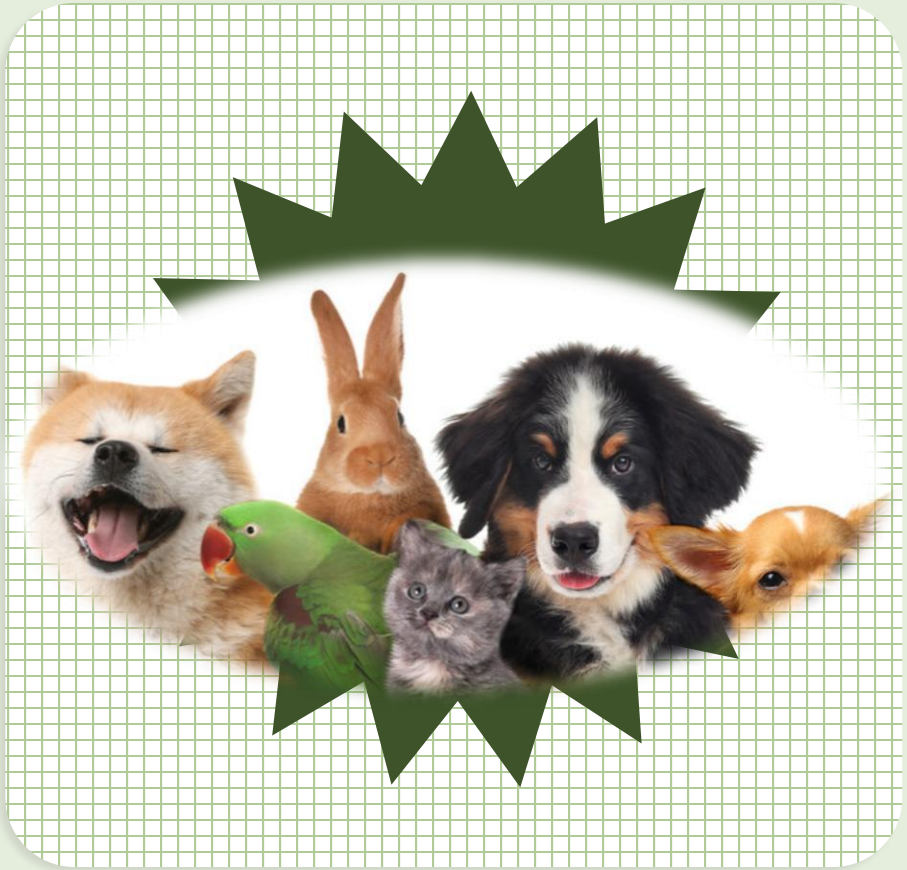
# Critical Path & Agile Response

## Technical Dependencies

- Gemini API – automated video tags
- Database/Cloud - video & data storage
- Data Pipeline – personalized feed

## Risk Mitigation

- Pet adoption used for harmful practices/bad environments
  - Implement background check and user screening
- 2+ users try to adopt the same pet simultaneously
  - Timer that locks availability of the pet the millisecond a user clicks to adopt
- Inaccurate representation of pet selection
  - Adoption pages must give additional information + potential call with candidate



# Thank you!!